MULTIPLICATION BY A UNIT ACTS ON THE DETERMINANT OF A FINITE LENGTH MODULE BY THE L-TH POWER OF THE UNIT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 02PD (sources/stacks-chow/pinned-original-source.tex line 16505).

Allowed standard prerequisites (editorial).

- The determinant of a finite length module and the additive isomorphism for a short exact sequence together with its equivariance. Same chapter, Lemma labelled lemma-det-exact-sequences (source line 16203), part (1), invoked at proof line 16527. Supplies the canonical isomorphism det(K) tensor det(M) -> det(L) and the property that it intertwines det(u_K) tensor det(u_M) with det(u_L), which is the relation f_K f_M = f_L used for the induction.
- The determinant functor on a finite length module over a local ring. Same chapter, chow.tex, the construction of det_kappa(M) in the periodic complexes section. Supplies the objects and the meaning of det_kappa(u_M) as an automorphism of a one dimensional kappa-vector space.

Source reference. Original lemma 02PA: sources/stacks-chow/pinned-original-source.tex, source lines 16203–16203. Same-chapter pointer into the author environment lemma at source line 16203; invoked at proof line 16527 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let R be a local ring with residue field κ. Let $u \in R^*$ be a unit. Let M be a module of finite length over R. Denote $u_M : M \to M$ the map multiplication by u. Then*

$$\det_\kappa(u_M) : \det_\kappa(M) \longrightarrow \det_\kappa(M)$$

is multiplication by $\overline{u}^l$ where $l = length_R(M)$ and $\overline{u} \in \kappa^*$ is the image of u.

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Proof. Denote $f_M \in \kappa^*$ the element such that $\det_\kappa(u_M) = f_M \mathrm{id}_{\det_\kappa(M)}$. Suppose that $0 \to K \to L \to M \to 0$ is a short exact sequence of finite R-modules. Then we see that u_k, u_L, u_M give an isomorphism of short exact sequences. Hence by Lemma 02PA (1) we conclude that $f_K f_M = f_L$. This means that by induction on length it suffices to prove the lemma in the case of length 1 where it is trivial. $\square$